

# Hongbin (Ben) Lin

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## Education

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### The Chinese University of Hong Kong

Ph.D. • Robotics • Mechanical and Automation Engineering

Nov 2020 - Mar 2026 • Hong Kong

- Advisor: Prof. Kwok Wai Samuel Au [Homepage](#)
- Research areas: world-model-based visuomotor learning, non-isometric equivariant data augmentation, robotic grasping, surgical autonomy, robotic system identification and gravity compensation

### The Chinese University of Hong Kong

M.Sc. • Mechanical and Automation Engineering

Sep 2017 - Jun 2018 • Hong Kong

### Guangdong University of Technology

B.S. • Mechanical and Automation Engineering

Sep 2013 - Jun 2017 • Guangzhou

## Work Experience

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### Robotics Manipulation Algorithm Engineer

Knowin.AI

Dec 2025 - Present • Shenzhen, China

(1) VLM-based visual perception, (2) robot trajectory planning, (3) compliant control, and (4) force sensing

### Senior Software Engineer

XDOF

Oct 2025 - Nov 2025 • Shenzhen, China

(1) HTC Vive-based data acquisition and (2) video encoding optimization

### Research Assistant (Advisor: Prof. Kwok Wai Samuel Au)

The Chinese University of Hong Kong

Jul 2018 - Oct 2020 • Hong Kong

## Publications

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1. Yonghao Long, Anran Lin, Derek Hang Chun Kwok, Lin Zhang, Zhenya Yang, Kejian Shi, Lei Song, Jiawei Fu, **Hongbin Lin**, Wang Wei, Kai Chen, Xiangyu Chu, Yang Hu, Hon Chi Yip, Philip Wai Yan Chiu, Peter Kazanzides, Russell H Taylor, Yunhui

- Liu, Zihan Chen, Zerui Wang, Kwok Wai Samuel Au, Qi Dou, "Surgical embodied intelligence for generalized task autonomy in laparoscopic robot-assisted surgery," *Science Robotics (SR)*, 2025. [\[PDF\]](#)
2. **Hongbin Lin**, Bin Li, Kwok Wai Samuel Au, "Visuomotor grasping with world models for surgical robots," *International Journal of Research and Review (IJRR)*, 2025, (under review). [\[PDF\]](#)
    - Proposed a world-model-based visuomotor learning framework for grasping tasks of surgical robots
  3. **Hongbin Lin**, Bin Li, Kwok Wai Samuel Au, "World models for general surgical grasping," *Robotics: Science and Systems (RSS)*, 2024. [\[PDF\]](#) [\[Project\]](#)
  4. **Hongbin Lin**, Bin Li, Xiangyu Chu, Qi Dou, Yunhui Liu, Kwok Wai Samuel Au, "End-to-end learning of deep visuomotor policy for needle picking," *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2023. [\[PDF\]](#) [\[Project\]](#)
  5. **Hongbin Lin**, Qian Gao, Xiangyu Chu, Qi Dou, Anton Deguet, Peter Kazanzides, Kwok Wai Samuel Au, "Learning deep nets for gravitational dynamics with unknown disturbance through physical knowledge distillation: initial feasibility study," *IEEE Robotics and Automation Letters (RAL)*, 2021 (presented in 2021 ICRA). [\[PDF\]](#) [\[Project\]](#)
  6. **Hongbin Lin**, Chiu-Wai Vincent Hui, Yan Wang, Anton Deguet, Peter Kazanzides, Kwok Wai Samuel Au, "A reliable gravity compensation control strategy for dVRK robotic arms with nonlinear disturbance forces," *IEEE Robotics and Automation Letters (RAL)*, 2019 (presented in 2019 IROS). [\[PDF\]](#) [\[Project\]](#)
    - Proposed a gravity compensation method for surgical manipulators with strong disturbances; the open-source package is used by 40+ institutions worldwide
  7. **Hongbin Lin**, Juan Rojas, Kwok Wai Samuel Au, "Multi-group equivariant augmentation for reinforcement learning in robot manipulation," *arxiv*, 2025. [\[PDF\]](#)
    - Proposed a non-isometric equivariant data augmentation method to improve the sampling efficiency of visuomotor learning
  8. Bin Li, Bo Lu, **Hongbin Lin**, Yaxiang Wang, Fangxun Zhong, Qi Dou, Yun-Hui Liu, "On the Monocular 3-D Pose Estimation for Arbitrary Shaped Needle in Dynamic Scenes: An Efficient Visual Learning and Geometry Modeling Approach," *IEEE Transactions on Medical Robotics and Bionics*, 2024. [\[PDF\]](#)
  9. Hongmin Wu, **Hongbin Lin**, Yisheng Guan, Kensuke Harada, Juan Rojas, "Robot introspection with bayesian nonparametric vector autoregressive hidden markov models," *2017 IEEE-RAS 17th International Conference on Humanoid Robotics (Humanoids)*, 2017. [\[PDF\]](#)
  10. Yan Wang, Hoi-Wut Yip, Hao Zheng, **Hongbin Lin**, Russell H Taylor, Kwok Wai Samuel Au, "Design and experimental validation of a miniaturized robotic tendon-driven articulated surgical drill for enhancing distal dexterity in minimally invasive spine fusion," *IEEE/ASME Transactions on Mechatronics (T-Mech)*, 2022. [\[PDF\]](#)
  11. Juan Rojas, Shuangqi Luo, Dingqiao Zhu, Yunlong Du, **Hongbin Lin**, Zhengjie Huang, Wenwei Kuang, Kensuke Harada, "Online robot introspection via wrench-based action grammars," *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017. [\[PDF\]](#)
  12. **Hongbin Lin**, Bin Li, Yunhui Liu, Kwok Wai Samuel Au, "Open-source high-precision autonomous suturing framework with visual guidance," *IEEE Int. Conf. Intell. Robots and Syst. (IROS) Workshop on "A Panacea Or An Alchemy? Benefits And Risks Of Robot Learning In Medical Applications"*, 2022. [\[PDF\]](#) [\[Project\]](#)

## Awards

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- Gold Award, the 9th China International College Students' "Internet+" Innovation and Entrepreneurship Competition [\[Media\]](#)
- Top Prize, the 17th "Challenge Cup" National College Students' Extracurricular Academic Science and Technology Contest [\[Media\]](#)

- Runner-up and Second Runner-up, Charles K. Kao Student Creativity Awards
- First Place, AccelNet Surgical Robotics Challenge

## Teaching

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- The Chinese University of Hong Kong · MAEG3080 Fundamentals of Machine Intelligence
- The Chinese University of Hong Kong · MAEG2050 Robot Development in Practice: From Design to Prototyping

## Skills

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### Profile Summary

A highly skilled researcher with over seven years of experience in robotics, specializing in independent algorithm development and the publication of research in top-tier journals. Demonstrates strong communication, collaboration, and presentation skills with a proven track record of delivering impactful results in multidisciplinary teams.

### Languages

Cantonese (Native), Mandarin (Fluent), English (Proficient)

### Programming

C/C++, MATLAB, Python

Updated: 2026.07.19